

Assignment 1 — Snowpipe Auto-Ingestion from GCS using Pub/Sub

Case Study: QuickCart Real-Time Order File Ingestion

QuickCart is an online retail company. Its order application generates small CSV files throughout the day. Every few minutes, a new order file is placed in a Google Cloud Storage bucket.

The analytics team currently loads these files manually into Snowflake. This causes delays and sometimes files are missed. Your task is to design and implement an automated ingestion pipeline so that a newly uploaded GCS file reaches a Snowflake table automatically.

Business Flow

Application → GCS → Pub/Sub notification → Snowpipe → Snowflake RAW table

Source Files

Create a folder/path in your GCS bucket for incoming order files.

orders_001.csv

order_id	customer_id	customer_name	product_category	order_amount	order_status	order_ts
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1001	C001	Ananya	Electronics	54000	PLACED	2026-08-12 09:10:00
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1002	C002	Ravi	Fashion	3200	PLACED	2026-08-12 09:15:00
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1003	C003	Meera	Home	7800	SHIPPED	2026-08-12 09:20:00
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orders_002.csv

order_id	customer_id	customer_name	product_category	order_amount	order_status	order_ts
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1004	C004	Arun	Electronics	22000	PLACED	2026-08-12 09:30:00
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1005	C001	Ananya	Home	4500	COMPLETED	2026-08-12 09:35:00
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1006	C005	Divya	Fashion	6800	CANCELLED	2026-08-12 09:40:00
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orders_003.csv

order_id	customer_id	customer_name	product_category	order_amount	order_status	order_ts
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1007	C006	Karthik	Electronics	18000	PLACED	2026-08-12 09:50:00
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1008	C002	Ravi	Home	9200	SHIPPED	2026-08-12 09:55:00
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1009	C007	Nisha	Fashion	5100	COMPLETED	2026-08-12 10:00:00
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Assignment Requirements

Implement the complete GCS-to-Snowflake automated ingestion flow. Your solution must demonstrate:

- A dedicated GCS bucket/path for incoming order files.
- Secure Snowflake access to GCS without embedding a Google service-account key in a stage.
- Identification of the Snowflake-generated identity that reads GCS.

- Least-privilege bucket permissions for that identity.
- A Snowflake external stage and CSV file format.
- A Pub/Sub topic for new-object notifications.
- A GCS object-finalize notification feeding the topic.
- A pull subscription.
- A Snowflake notification integration connected to the subscription.
- Identification of the Snowflake-generated Pub/Sub service account and the required permissions.
- A Snowpipe configured for automatic ingestion.
- Uploading the files one at a time and proving they are loaded without manually running a COPY command for each file.
- Validation that normal load-history behavior does not duplicate an already loaded file.
- Pipe-status and load-history checks for troubleshooting.

Important Architecture Question

Snowflake uses cloud identities for two different purposes in this assignment:

1. Reading files from GCS.
2. Reading event messages from Pub/Sub.

Identify both identities and explain why both are required.

Validation Checkpoints

- After orders_001.csv: 3 rows.
- After orders_002.csv: 6 rows.
- After orders_003.csv: 9 rows.
- A previously loaded file should not normally create duplicate rows just because the pipeline checks the location again.

Deliverables

Submit an architecture diagram, GCS screenshots, Pub/Sub screenshots, IAM screenshots, Snowflake object details, final target-table result, pipe status/load history, and a short troubleshooting note.

Restriction

Do not solve the assignment by manually loading every file. The objective is event-driven Snowpipe auto-ingestion.

Note :use meaningful naming convention